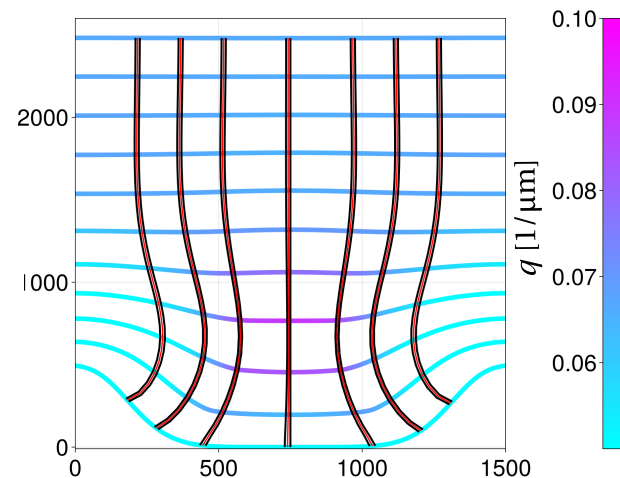
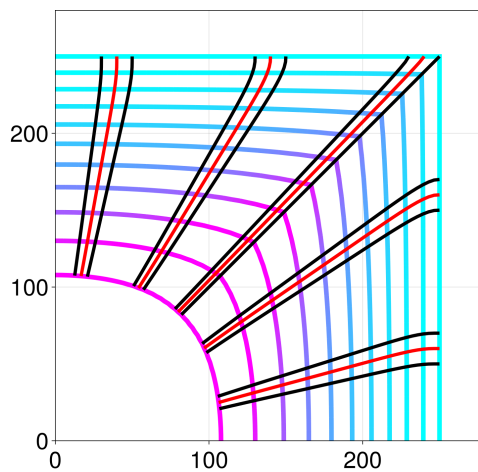
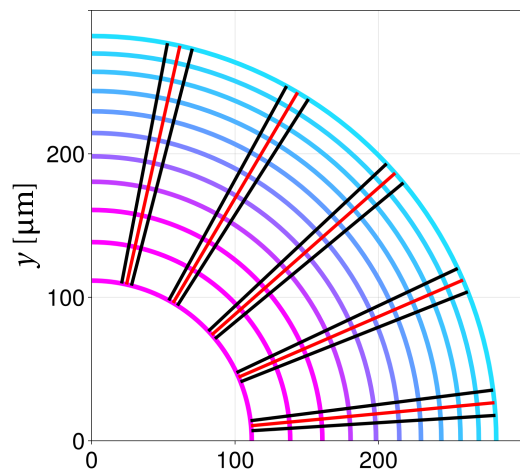


a)

Linear force: $f(\ell) = k(\ell - a)$ 

b)

Nonlinear force: $f(\ell) = ka^2\left(\frac{1}{a} - \frac{1}{\ell}\right)$ 